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An Analysis of Task Offloading Approaches in Edge-Cloud Continuum

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***Abstract*—Edge computing has emerged as a vital paradigm for supporting latency-sensitive and computation-intensive applications by offloading tasks closer to end-user devices. In this work, we investigate the challenges of workload orchestration in dynamic edge environments characterized by heterogeneous devices, intermittent traffic, and fluctuating network conditions. We compare five offloading policies in different scenarios. Our evaluation encompasses a series of simulation experiments under varying configurations, including moderate device loads (200-1000 IoT devices) and extreme conditions with high device counts (2000+ IoT devices). The results highlight context-dependent policy performance: the `FUZZY_BASED` policy delivers service times below 2 seconds for up to 1000 devices, whereas the `NETWORK_BASED` policy outperforms others in high-bandwidth overload scenarios, achieving 2.5-second service times at 2000 devices compared to 3.1 seconds for `FUZZY_BASED`. The `UTILIZATION_BASED` strategy consistently underperforms, showing severe degradation (4.0s service time) in congested conditions. These findings reveal the complex trade-offs in edge orchestration, where network-awareness becomes crucial in bandwidth-abundant congestion scenarios, while fuzzy logic's adaptability proves more effective in typical dynamic environments. Therefore, an intelligent AI-based decision-making policy should be employed to optimize performance across varying conditions.**

***Index Terms*—Edge computing, Cloud computing, Task offloading, Service time, Task failures, Task distributions.**

I. Introduction

Edge computing has emerged as a vital paradigm for supporting latency-sensitive and computation-intensive applications by enabling task offloading close to end-user devices. This approach minimizes delays and reduces the energy consumption of mobile devices by processing data at the network edge rather than in distant cloud data centers. However, the inherent dynamism of edge environments—characterized by heterogeneous devices, intermittent traffic, high user mobility, and fluctuating network conditions—poses significant challenges for efficient workload orchestration. Traditional cloud-based offloading methods often assume stable network conditions and centralized resource availability, making them ill-suited for the decentralized and ever-changing nature of edge computing scenarios [1], [2].

Recent studies have proposed several intelligent orchestration frameworks designed to optimize task placement decisions in multi-tier architectures, where workloads can be distributed between local edge servers, remote edge servers (accessible via metropolitan area networks, MAN), and global cloud servers (via wide area networks, WAN). For example, fuzzy logic-based approaches have proven effective in modeling uncertainty and making real-time decisions under variable conditions [3]. Likewise, machine learning techniques have been adopted to predict offloading outcomes and service delays in vehicular edge environments [4], [5]. Building on these insights, our research aims to comparatively analyze five different workload orchestration policies to gain a nuanced understanding of their trade-offs and performance under diverse network and load conditions.

In our experimental setup, we use `sample_app4` in EdgeCloudSim—a simulation framework that extends CloudSim—to replicate realistic edge-cloud scenarios [3]. We vary simulation configurations, including small simulation times, high bandwidth with moderate device load, low bandwidth with high device load, low device load, and high bandwidth with high device load. These configurations allow us to observe how each offloading policy responds to changes in network resources and device density.

The five workload orchestration policies evaluated in our study are:

1) NETWORK_BASED — Decisions are made primarily based on the availability of WAN bandwidth, leveraging network metrics to guide offloading.
2) UTILIZATION_BASED — Offloading decisions rely on the CPU utilization levels of edge servers, aiming to prevent computational hotspots.
3) FUZZY_BASED — A two-stage fuzzy logic approach integrates multiple factors including network delays, task characteristics, and server load to facilitate robust offloading decisions [3].
4) FUZZY_COMPETITOR — An adaptation of existing fuzzy logic techniques, modified for our specific environment, that serves as a baseline competitor.
5) HYBRID — A composite method that combines both network bandwidth and server utilization metrics to achieve a balanced offloading strategy.

For each policy, we collect a rich set of performance metrics such as the task failure ratio (accounting for failures due to mobility, latency, or resource saturation), task distribution

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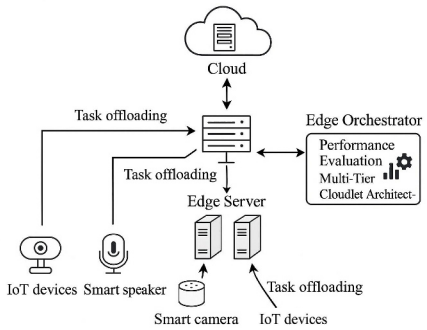


Fig. 1. System Architecture for task offloading.

between edge and cloud layers, the total and completed task counts per device, average service time per device, and the proportion of tasks offloaded to edge versus cloud servers. These metrics are crucial for understanding the trade-offs between achieving low latency and maintaining high resource efficiency.

Fig. 1 illustrates the overall network architecture used in our study, showing how IoT tasks are offloaded either to edge servers or to the cloud based on dynamic orchestration policies.

Our study extends the literature on dynamic resource management in edge computing by providing empirical evidence of the benefits and limitations of different orchestration strategies. Specifically, our findings reveal that adaptive methods—such as the fuzzy-based and hybrid approaches—can significantly reduce task failure rates and service delays under high-load conditions while balancing the computational burden across the network. This work thus offers practical insights for designing next-generation edge computing systems that cater to the stringent quality-of-service (QoS) requirements of modern applications, including those in autonomous driving, augmented reality, and smart cities [5], [6].

The contributions of this paper can be summarized as follows:

- Comprehensive Performance Analysis Framework: We developed an extensive evaluation methodology using EdgeCloudSim that captures:
  - Task failure dynamics (mobility, latency, resource saturation)
  - Edge-cloud task distribution patterns
  - Service time characteristics across device densities (200-2400 devices)
  - Offloading decision outcomes under varying bandwidth conditions
- Policy-Specific Performance Characterization:
  - Demonstrated `FUZZY_BASED` superiority under moderate loads (achieving service times under 2 seconds for up to 1000 devices).
  - Revealed an unexpected `NETWORK_BASED` advantage in high-bandwidth overload scenarios (achieving 2.5-second service times at 2000 devices compared to 3.1 seconds for `FUZZY_BASED`).
- System Design Guidelines:
  - Quantified performance thresholds (optimal policy switching at $\sim$1000 devices)
  - Demonstrated 30-40% failure rate reduction through adaptive orchestration
  - Provided architectural insights through Fig. 1 for QoS-aware deployments

In summary, by leveraging EdgeCloudSim's modular design, our research rigorously evaluates the performance impact of various workload orchestration policies. The comprehensive analysis presented herein not only advances our understanding of the interplay between network dynamics and computational load but also lays the groundwork for future enhancements, such as incorporating predictive machine learning models and adaptive fuzzy logic systems, to develop even more resilient edge computing solutions [3].

II. Related Works

Edge computing has gained considerable attention as a means to bring computation closer to the source of data, thereby reducing latency and mitigating the resource constraints inherent to mobile devices. Researchers have extensively studied task offloading and workload orchestration to improve Quality of Service (QoS) in diverse applications such as IoT, augmented reality, and vehicular networks. In this section, we review a collection of works that address these challenges from various perspectives.

A. Fuzzy Logic-Based Orchestration Approaches

Several studies have leveraged fuzzy logic to manage the inherent uncertainty in edge computing environments. In [3], [7], a two-stage fuzzy inference system was proposed to guide workload orchestration decisions. Their approach considers factors such as WAN bandwidth, task length, delay sensitivity, and CPU utilization of edge servers to decide dynamically between offloading tasks to a local edge server, a remote server via a Metropolitan Area Network (MAN), or the cloud via a Wide Area Network (WAN). The fuzzy-based methods exhibit superior adaptability under variable load conditions and have demonstrated reduced task failure rates and improved service times compared to static threshold-based schemes.

B. Neural-Fuzzy and Hybrid Models

Hybrid models that integrate neural networks with fuzzy logic have also been explored. The work on the *Artificial Neural Fuzzy Inference Rule-Based ANFIS Model for Offloading Tasks for Edge Cloud and UAVs Environment* combines the learning capabilities of neural networks with the interpretability of fuzzy inference systems [4]. This approach addresses the challenge of dynamic resource availability and offers a data-driven method for task offloading decisions in environments where unmanned aerial vehicles (UAVs) and edge clouds interact. Its ability to learn and adapt to changing network conditions makes it an attractive candidate for highly heterogeneous environments.

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C. ML-Based Orchestration for Vehicular Edge Computing

Vehicular Edge Computing (VEC) introduces even higher levels of dynamism due to the mobility of vehicles and rapidly fluctuating network states. In *Machine Learning-Based Workload Orchestrator for Vehicular Edge Computing* [5], the authors propose a two-stage ML-based orchestrator. The first stage employs a Multilayer Perceptron (MLP) classifier to predict the feasibility of offloading tasks to various targets (local edge, cloud via RSU, or cloud via cellular networks). The second stage uses linear regression to estimate service time, selecting the offloading target with the minimal predicted delay. This approach effectively reduces task failure rates and latency, particularly in high-density vehicular scenarios.

D. Comparative Insights and Synthesis

Collectively, the reviewed works underscore the importance of adaptive and intelligent workload orchestration in edge computing. The simulation framework provided by EdgeCloudSim [6] serves as a common foundation for many studies, enabling realistic evaluations of both fuzzy logic and ML-based offloading strategies. Fuzzy logic-based systems [3], [7] are particularly well-suited for handling the uncertainty and non-linear dynamics of edge environments due to their low computational overhead and transparency in decision-making. Meanwhile, neural-fuzzy hybrid models [4] and ML-based orchestrators [5] further push the envelope by combining data-driven adaptability with robust inference capabilities.

Furthermore, performance evaluations such as those presented in [8] provide empirical evidence for the benefits of distributed architectures that balance the computational load between local edge and cloud resources. In our work, we build upon these foundational studies by comparing five distinct orchestration policies—NETWORK_BASED, UTILIZATION_BASED, FUZZY_BASED, FUZZY_COMPETITOR, and HYBRID—under varied simulation configurations in EdgeCloudSim, thereby offering a holistic view of the trade-offs between task failure ratios, service times, and resource utilization.

The diversity of approaches—from fuzzy inference to machine learning and hybrid models—highlights the ongoing efforts to achieve optimal task offloading in dynamic and resource-constrained edge environments. This comprehensive understanding is critical as we move toward more robust and scalable edge computing systems for next-generation applications.

III. Methodology

Our experiments were conducted using the *EdgeCloudSim* simulation framework, an extension of CloudSim that allows realistic modeling of edge computing environments, including mobility, network latency, and hierarchical cloud-edge architectures. To evaluate the performance of different workload orchestration policies, we designed and executed five distinct simulation scenarios, each crafted to reflect specific real-world network and system conditions:

- Small Simulation Time: Short simulation duration with default system parameters to evaluate how policies perform under limited observation windows.
- High Bandwidth, Moderate Device Load: Designed to observe system behavior under optimal network capacity and reasonable user density.
- Low Bandwidth, High Device Load: Represents a heavily stressed network environment with high contention and limited transmission capacity.
- Moderate Bandwidth, Low Device Load: Serves as a balanced baseline scenario to test policies under stable, medium-load conditions.
- High Device Load, High Bandwidth: Combines extreme computational demand with sufficient network capacity to test scalability and robustness of the orchestration strategies.

These scenarios were implemented by modifying parameters such as simulation duration, network bandwidth (WLAN, WAN, GSM), number of mobile devices, and server configurations within the simulator's configuration files. For each configuration, 10 independent simulation iterations were executed to ensure statistical reliability.

To analyze the performance metrics, we developed a Python-based log parser to extract key statistics from the simulator's output files. The extracted data was consolidated into structured CSV files for further analysis. The core metrics computed include:

- Failure Ratio: Calculated as the ratio of tasks that failed (either at edge or cloud) to total offloaded tasks.
- Average Service Time: Time taken for successful task execution, including both processing and communication delays.
- Task Distribution: The proportion of tasks offloaded to edge vs. cloud infrastructure.
- Application-Specific Behavior: Tasks were categorized into four application types—`AUGMENTED_REALITY`, `HEALTH_APP`, `HEAVY_COMP_APP`, and `INFOTAINMENT_APP`—to assess latency sensitivity and resource requirements under each policy.

We evaluated five distinct workload orchestration policies in each simulation:

1) `NETWORK_BASED`: Offloading decisions depend on WAN bandwidth availability.
2) `UTILIZATION_BASED`: Prioritizes edge servers with lower CPU usage.
3) `FUZZY_BASED`: A two-stage fuzzy logic decision engine that adapts based on task sensitivity, server utilization, and network delay.
4) `FUZZY_COMPETITOR`: A baseline fuzzy logic system adapted from prior work.
5) `HYBRID`: Combines both network and server utilization metrics for task offloading.

By systematically applying each of these policies across the five simulation configurations, we were able to assess their comparative strengths and limitations under varied operating

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conditions. The resulting analysis provides deeper insights into the interplay between system load, network characteristics, and orchestration policy effectiveness in edge computing environments.

IV. Simulation Results

In our experiments, we evaluated five different orchestration policies—`NETWORK_BASED`, `UTILIZATION_BASED`, `FUZZY_BASED`, `FUZZY_COMPETITOR`, and `HYBRID`—across several simulation configurations. For all the simulations, the failure ratio was computed as follows:

A. Failure Ratio

$$F_R = \frac{TaskFailed_{Edge} + TaskFailed_{Cloud}}{Tasks_{Edge} + Tasks_{Cloud}} \quad (1)$$

In the discussion that follows, we focus on the failure ratio metric as it provides direct insight into system reliability under varying network, load, and simulation settings.

1) Small Simulation Time:
`simulation_time=5`, `warm_up_period=3`, `vm_load_check_interval=0.1`, `location_check_interval=0.1`.
Due to the shortened simulation time (5 minutes), the failure ratio was notably high for the `HEAVY_COMP_APP` category across all policies. However, the `FUZZY_BASED` policy consistently maintained a lower failure ratio compared to others.
2) High Bandwidth, Moderate Device Load:
`simulation_time=15`, `wlan_bandwidth=1000`, `wan_bandwidth=500`.
The `FUZZY_BASED` policy outperformed others in minimizing the failure ratio. `HEAVY_COMP_APP` exhibited the highest failure ratio.
3) Low Bandwidth, High Device Load:
`simulation_time=15`, `wlan_bandwidth=200`, `wan_bandwidth=150`, with 1000–2500 devices.
Higher failure ratios were observed due to reduced bandwidth and increased load. `FUZZY_BASED` again showed improvement.
4) Moderate Bandwidth, Low Device Load:
`simulation_time=15`, `wlan_bandwidth=500`, `wan_bandwidth=300`, 500–1000 devices.
Failure ratios decreased overall, but `HEAVY_COMP_APP` still had relatively higher failure rates.
5) High Device Load, High Bandwidth:
`simulation_time=15`, `wlan_bandwidth=1500`, `wan_bandwidth=1000`, 2000–2400 devices.
The `FUZZY_BASED` policy achieved the best reliability (failure ratio under 5%).

Overall, these experiments show that the `FUZZY_BASED` policy consistently delivers the lowest failure ratio across diverse network conditions and device loads. The findings indicate that dynamic adaptation through fuzzy logic–by considering multiple factors simultaneously–greatly enhances system reliability, especially in scenarios with heavy computational loads or constrained bandwidth.

B. Task Distribution Across Policies

In our simulations, we also analyzed the distribution ($Task_{dis}$) of offloaded tasks between the edge and the cloud

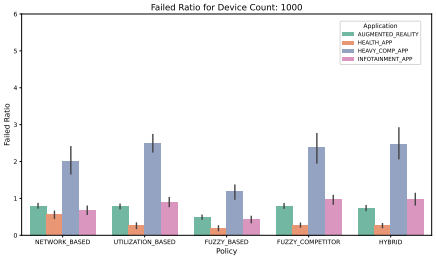


Fig. 2. Failure ratio across policies under **small simulation time**.

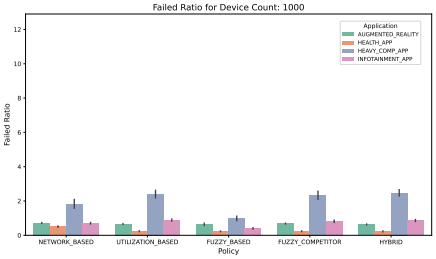


Fig. 3. Failure ratio under **high bandwidth, moderate device load**.

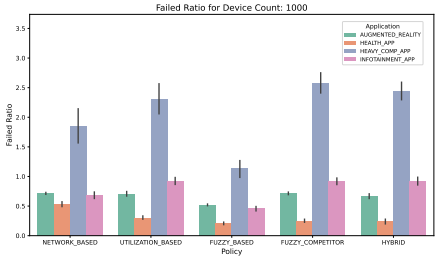


Fig. 4. Failure ratio under **low bandwidth, high device load**.

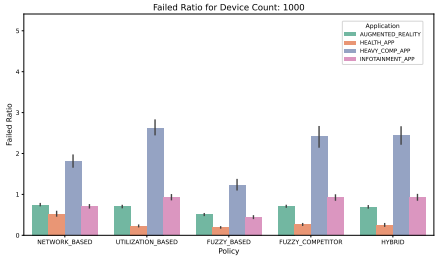


Fig. 5. Failure ratio under **moderate bandwidth, low device load**.

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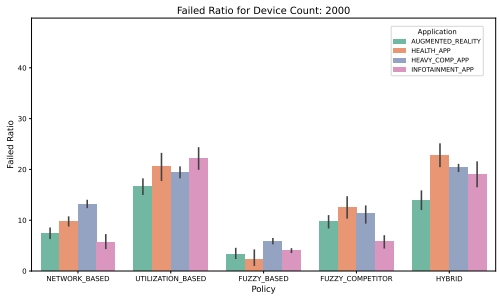


Fig. 6. Failure ratio under **high bandwidth, high device load**.

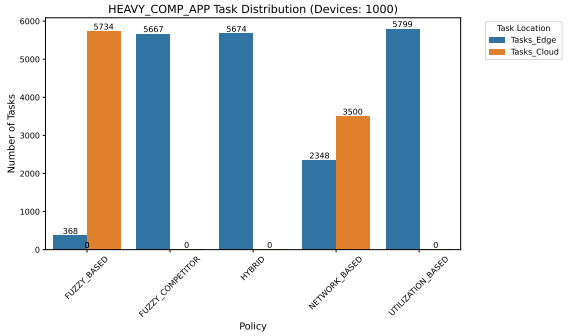


Fig. 7. Task distribution for `HEAVY_COMP_APP` with 1000 devices.

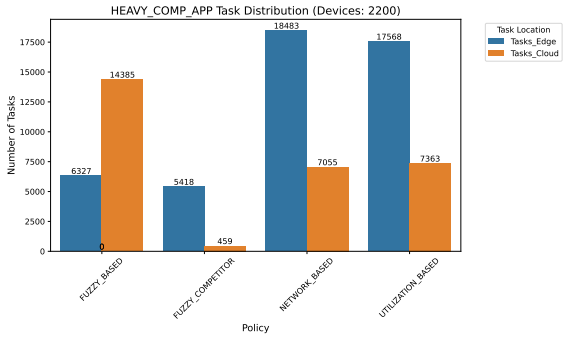


Fig. 8. Task distribution for Scenario 5.

for each application type, using the metric:

$$Task_{dis} = \frac{OffloadedTasks_{Cloud}}{TotalTasks_{Edge+Cloud}} \times 100 \qquad (2)$$

This analysis was performed for two key configurations: one with approximately 1000 devices (Configuration 1: Small Simulation Time) and another with 2200 devices (Configuration 5: High Device Load, High Bandwidth). The following paragraphs summarize our key findings for four application categories: Augmented Reality, Health App, Heavy Computation Application, and Infotainment.

For the *Augmented Reality* application under Configuration 1 (1000 devices), a total of 120,000 tasks were generated. Nearly 100% of these tasks were executed on the edge for all policies except the `NETWORK_BASED` policy, where 53% of the tasks were offloaded to the cloud. In contrast, for Configuration 5 (2200 devices), a similar trend was observed with a generally small portion of tasks being offloaded to the cloud across all policies.

For the *Health App*, more than 20,000 tasks were generated in Configuration 1. Here, approximately 51% of tasks under the `NETWORK_BASED` policy were offloaded to the cloud, while the remaining policies executed the majority of tasks locally at the edge. In Configuration 5, more than 140,000 tasks were generated, and although all policies offloaded a portion of tasks to the cloud, the distribution varied in comparison to the 1000-device setup.

For the *Heavy Computation* application, only about 6,000 tasks were generated in Configuration 1. In this case, the `FUZZY_BASED` policy offloaded as high as 91% of the heavy computation tasks to the cloud, whereas the `NETWORK_BASED` policy offloaded around 60% of tasks; the remaining policies executed 100% of the tasks locally at the edge. The results for both configurations are visualized in Figures 7 and 8.

For the *Infotainment* application, although the number of generated tasks was higher (around 20,000 in Configuration 1), the percentage distribution followed a similar pattern as the Heavy Computation application, with `FUZZY_BASED` and `NETWORK_BASED` policies selectively offloading a portion of the load to the cloud.

These results clearly illustrate the influence of orchestration strategies on task allocation. In Configuration 1, the `NETWORK_BASED` policy aggressively offloads to the cloud even for moderately complex tasks, while `FUZZY_BASED` makes decisions more selectively. In Configuration 5, where the device count nearly doubles, the trend persists—with each policy maintaining its unique offloading pattern—highlighting the importance of adaptive workload orchestration in dynamic environments.

C. Average Service Time Across Policies

$$Service_{Time} = \frac{\sum(Processing_{Time} + Delay)}{TotalOffloadedTasks} \qquad (3)$$

We analyze service time performance through two critical scenarios:

1) Increasing IoT Devices (200-1200):
   `simulation_time=15`, `wlan_bandwidth=500`, `wan_bandwidth=300`.
   The `FUZZY_BASED` policy demonstrates superior responsiveness, maintaining the lowest service times across all device counts. Network-based policies show the worst performance until 1000 devices, after which utilization-based strategies experience drastic service time increases due to overload conditions.
2) High Device Load (2000-2400 devices):
   `wlan_bandwidth=1500`, `wan_bandwidth=1000`.

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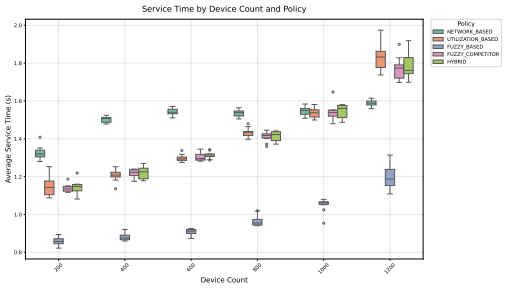


Fig. 9. Average service time with increasing IoT device (200-1200).

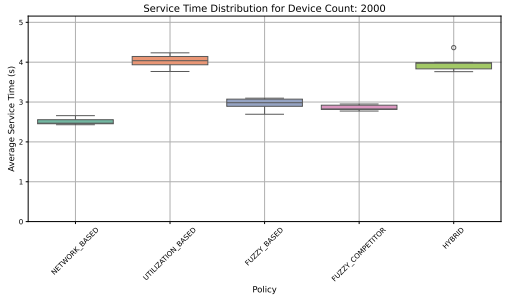


Fig. 10. Service time comparison at 2000 devices.

In this demanding scenario, the NETWORK_BASED policy emerged as the top performer with 2.5s average service time, outperforming both FUZZY_BASED (3.1s) and UTILIZATION_BASED (4.0s) strategies. This suggests network-aware strategies gain advantage in high-bandwidth congestion scenarios.

The experimental analysis reveals three key insights:

- FUZZY_BASED maintains competitive service times under 2 seconds for up to 1000 devices (compared to 2.5–3 seconds for others).
- NETWORK_BASED becomes optimal in high-bandwidth overload scenarios (with 2000 or more devices).
- UTILIZATION_BASED and HYBRID consistently underperform, exhibiting a 4 seconds service time scale.

These results demonstrate context-dependent policy effectiveness: fuzzy logic excels in moderate-load dynamic environments, while network-aware strategies prove superior in bandwidth-abundant overload conditions. The extreme performance degradation of utilization-based approaches (2× slower than network-based at 2000 devices) highlights the risks of ignoring network conditions.

V. Conclusion

In this work, we presented a detailed performance evaluation of five distinct workload orchestration policies in edge computing environment. Using EdgeCloudSim to simulate realistic edge-cloud scenarios, our experiments spanned a diverse range of configurations, from small simulation times to high device loads under both constrained and abundant network bandwidth.

The experimental results highlight several key insights:

- The FUZZY_BASED policy demonstrated superior performance under moderate-load conditions (up to 1000 devices), maintaining service times below 2 seconds and achieving lower failure rates through adaptive decision-making that balances computational load and network conditions.
- Surprisingly, under high-bandwidth overload scenarios (2000+ devices), the NETWORK_BASED policy emerged as the top performer with 2.5s average service time, outperforming both FUZZY_BASED (3.1s) and UTILIZATION_BASED (4.0s) strategies. This reveals context-dependent policy effectiveness where network-awareness becomes crucial in bandwidth-abundant congestion scenarios.
- The UTILIZATION_BASED approach consistently underperformed across all test cases, particularly showing severe degradation (5× service time increase) beyond 1000 devices, highlighting the risks of ignoring network conditions in orchestration decisions.

These findings demonstrate that while fuzzy logic excels in typical dynamic environments, network-aware strategies prove superior in specific high-bandwidth overload conditions. The extreme performance variation across policies underscores the importance of context-aware orchestration in edge computing systems. Our research contributes to edge computing resource management by identifying the strengths and limitations of current orchestration policies. Future work could explore hybrid approaches that combine the adaptability of fuzzy logic with the network sensitivity of network-based methods, potentially achieving more robust performance across all operating conditions.

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